

Early Exit Deep Learning: A Pathway to Sustainable AI through Reduced Computational Footprint

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Abstract: The proliferation of deep neural networks across real-world applications has intensified concerns regarding computational efficiency and energy consumption, particularly in resource-constrained deployment environments. Uniform full-depth inference — where every input traverses all layers regardless of its classification difficulty — represents a fundamental inefficiency in standard deep learning pipelines. This paper presents a comprehensive empirical study of confidence-based early-exit inference across four distinct neural architectures applied to the CIFAR-10 benchmark: a custom three-block convolutional neural network (CNN), a pretrained ResNet-18 with partial fine-tuning, a stacked long short-term memory network (LSTM), and a pretrained Vision Transformer (ViT-B/16). Each architecture is augmented with lightweight exit heads at three intermediate points, and inference is dynamically terminated when the softmax maximum probability exceeds a tunable confidence threshold, ranging from 0.50 to 0.99. All models are trained end-to-end using joint auxiliary loss optimization with exit-weighted coefficients of 0.5, 0.3, and 0.2 respectively, and evaluated via deterministic floating-point operation (FLOP) counting as an energy consumption proxy.

Experimental results reveal that architectural design and pretraining quality are the dominant determinants of early-exit effectiveness, surpassing the influence of the exit mechanism itself. For the pretrained ViT-B/16, we obtain an accuracy of 97.5% with 69% of test data being processed through the first exit point when the threshold is set to 0.80, obtaining a maximum of 18.8% FLOPs saving while maintaining less than 0.6% loss of accuracy. On the other hand, the pretrained ResNet-18 obtains an accuracy of 92.5%, but cannot exploit the early-exits approach owing to the use of frozen intermediate layers that produce inaccurate confidence values, saving only 2% of FLOPs. The custom CNN demonstrates the clearest Pareto-optimal trade-off

curve, reducing average inference FLOPs by up to 35% at threshold 0.50 with a 5.1% accuracy penalty. The LSTM, which serializes image rows as sequential time steps, plateaus at approximately 60% accuracy regardless of threshold, confirming that early-exit confidence never materializes when the underlying model representations are architecturally mismatched to the input domain.

These findings contribute actionable insights for Green AI practitioners: confidence-based early exiting is most effective when combined with strong pretrained representations that naturally produce high-confidence intermediate activations, and frozen transfer learning strategies are contraindicated for early-exit deployment.

Keywords: *early-exit networks, confidence-based inference, Green AI, energy-efficient deep learning, Vision Transformer, ResNet, LSTM, convolutional neural networks, FLOP analysis, CIFAR-10, dynamic inference. Green AI, confidence-based inference.*

1. INTRODUCTION

Deep neural networks (DNNs) provide state-of-the-art results across various problems but impose the same computation across all instances regardless of their hardness, leading to wasted computation on easy instances and increasing carbon emissions as AI applications scale up. This is against the principles of Green AI, which seeks energy-efficient inference that aligns with the problem hardness, especially in scenarios where limited computational resources exist at the edge or when scaling up to clouds. Early exit models solve this issue by introducing side classifiers in intermediate layers to allow for early termination of easy instances.

Previous research on early exit networks has largely considered only one network architecture, such as ResNet or bespoke CNNs, and has neglected to

consider how architectural decisions like representation quality affect the performance of confidence-based exits. This omission is both practical and meaningful because an engineer deciding between training a fresh CNN from scratch, fine-tuning ResNet only partially, using an RNN, or employing a transformer network with pre-trained representations will realize very different efficiency benefits from including early exits.

This paper makes the following contributions:

- A systematic empirical study of early-exit inference across four architecturally diverse models on CIFAR-10, using identical exit head designs and threshold sweep protocols for controlled comparison.
- A replicable training framework with joint auxiliary loss training (exit weight factors: 0.5, 0.3, 0.2), deterministic estimation of number of Floating Point Operations, and threshold variations between 0.50 and 0.99 for Pareto frontier discovery.
- Proof that the quality of pretraining and calibration of representations, rather than the exit strategy itself, determine early-exit performance.
- A student-accessible framework that bridges the gap between complex industrial early-exit systems and interpretable, low-overhead mechanisms suitable for sustainable ML research.

2. LITERATURE REVIEW

Deep neural networks are accurate but consume excess computational energy, making them inefficient due to their full-depth pass-through of all inputs, and thus prompting research into efficient models such as early exit strategies. The BranchyNet framework proposed by Teerapittayanon et al. (2017) is one of the earliest attempts at a multi-exit deep learning model for efficient inference. But despite the potential benefits of using such an architecture, this paper lacks any evaluation in terms of FLOPs or energy consumption.

Kaya et al. (2019) introduced Shallow-Deep Networks (SDN), inserting internal classifiers into VGG, ResNet, WideResNet, and MobileNet architectures on CIFAR-10, CIFAR-100, and Tiny-ImageNet, achieving up to 75% inference cost reduction. Critically, they identified the 'overthinking' phenomenon — where deeper layers sometimes reverse correct shallow predictions —

motivating confidence-gated termination. Our work builds on this framing but extends it to architecturally heterogeneous settings including recurrent and transformer models.

Demir and Akbas (2024) proposed EENet with confidence and softmax branches combined with a loss function balancing accuracy and cost, reducing computational cost to 20-42% of baseline with minimal accuracy drop across MNIST, SVHN, CIFAR-10, and Tiny-ImageNet. Jo et al. (2023) demonstrated hardware-aware early-exit design with Dynamic Branch Pruning in LoCoExNet, achieving 91% energy savings and 4.46x speedup on CIFAR-10 with VGG-16 on a 65nm CMOS accelerator. These hardware-co-designed approaches achieve impressive results but are not reproducible without specialized hardware.

Mokssit et al. (2026) introduced Confidence-Gated Training (CGT), which aligns training-time gradients with inference-time exit policies, improving early-exit accuracy by reducing the mismatch between joint-loss training objectives and threshold-driven inference. This directly motivates our use of weighted auxiliary losses in training. Liang et al. (2023) applied temporal early-exit strategies to spiking neural networks (SNNs), achieving same accuracy with fewer timesteps; while not directly comparable to our CNN/ViT study, it establishes that early-exit principles generalize across computation paradigms.

Recent work by Li et al. (2023) explored learned exit policies via self-distillation in adaptive early-exit scheduling, achieving higher accuracy at the same latency budget. Schneider et al. (2024) applied neural architecture search (NAS) to jointly optimize backbone and exit head placement, improving the accuracy-MAC trade-off but at high design complexity. A comprehensive survey by Rahmath et al. (2024) taxonomized early-exit strategies across architecture, training, and exit policy dimensions, explicitly highlighting the lack of threshold-energy trade-off analysis as a key gap — the gap our work directly addresses through systematic Pareto curve generation.

Table 1. Literature Survey: Early-Exit Deep Learning Methods — Techniques, Datasets, Results, and Research Gaps

Reference	Techniques Used	Dataset / Algorithms Used	Results (Key Finding)	Research Gap / Future Opportunity
Kaya et al., 2019, ICML (Shallow-Deep Networks)	Internal classifiers (SDN) with confidence-based early exits	CIFAR-10, CIFAR-100, Tiny-ImageNet; VGG, ResNet, WideResNet, MobileNet	Up to 75% inference cost reduction with maintained or improved accuracy; overthinking identified as a major inefficiency	Expand to other architectures and real-time applications; combine with confusion-aware metrics
Demir & Akbas, 2024, arXiv (EENet)	Early-exit CNN with confidence and softmax branches; loss function balancing accuracy and cost	MNIST, SVHN, CIFAR-10, Tiny-ImageNet; ResNet backbone	Computational cost reduced to 20–42% of baseline with minimal accuracy drop	Robust early-exit strategies under distribution shift or adversarial inputs
Mokssit et al., 2026, arXiv (CGT)	Confidence-Gated Training (CGT); sample-dependent gradient control for early exits	Fashion-MNIST, Indian Pines	Improved early-exit accuracy and reduced overthinking; aligns training with inference policy	Integrate CGT with large-scale models and benchmark on real-time edge hardware
Jo et al., 2023, IEEE TCAD (LoCoExNet)	Low-cost early exits + Dynamic Branch Pruning (DBP); hardware-aware design	Tiny-ImageNet, CIFAR-10; VGG-16; 65nm CMOS accelerator	91% energy savings, 4.46× speedup without accuracy loss	Deploy on diverse hardware setups; test with dynamic workloads
Liang et al., SEENN, NeurIPS, 2023	Temporal early exit (SNN)	Confidence timestep exit, RL policy	Same accuracy with fewer timesteps	SNN-only; not CNN energy study
Schneider et al., AEBNAS, IEEE, 2024	NAS-based early-exit design	Hardware-aware NAS, margin exit threshold	Better accuracy–MAC trade-off	High complexity; not interpretable
Teerapittayanon et al., BranchyNet, IEEE, 2017	Early-exit CNN with side branches	Multi-exit architecture, entropy-based exit, joint training	Up to 5× speedup with similar accuracy	No FLOPs/energy analysis; fixed thresholds
Li et al., EENet, IEEE, 2023	Adaptive early-exit scheduling	Multi-exit DNN, self-distillation, learned exit policy	Higher accuracy at same latency	Complex policy; limited energy study
Khan et al., Hybrid Early-Exit, IEEE, 2020	Quantized early-exit + edge-cloud	Hybrid CNN, entropy exit, MAC energy model	~5× energy reduction	Hardware-specific; no FLOPs trade-off
Rahmath et al., Early-Exit Survey, ACM, 2024	Survey & taxonomy	Architecture, training, exit policy classification	Identifies efficiency benefits	Highlights lack of threshold–energy analysis

3. PROBLEM STATEMENT

The objective of this study is to develop and empirically evaluate confidence-based early-exit inference across four distinct neural architectures on the CIFAR-10 classification benchmark, producing quantitative accuracy-energy trade-off curves and isolating the architectural and training factors that determine early-exit effectiveness.

Formally, given an input image x and a neural network partitioned into K sequential blocks $f_1, \dots, f_K$, each followed by a lightweight exit head e_k , inference at test time proceeds as follows: after each block k , exit head e_k computes a softmax distribution over $C=10$ classes. Inference stops and the prediction is made if $\max_c p_k(c|x) \geq \tau$ for some user-defined threshold τ between $[0.50, 0.99]$. Otherwise, inference proceeds to block $k+1$. In the absence of an exit decision from any of the blocks up to $K-1$, the last classifier at block K is used.

Our objective is to optimize for maximal accuracy and minimal average inference FLOPs on the test set. We select operating thresholds that provide a significant amount of energy savings on easily classified inputs while ensuring that accuracy is not significantly compromised on difficult inputs.

For reporting results, we present accuracy and average FLOPs per sample (in comparison to the baseline full-depth FLOPs). In addition, we include the exit rate per each block and the percentage reduction in FLOPs compared to full-depth inference.

An important consideration is reproducibility. All experiments were done with deterministic random seeds and all models are trained with fixed seeds. FLOP counts were calculated analytically with ptflops. No measurement-based wall-clock time was collected, and threshold sweeps were consistent across all four different model types.

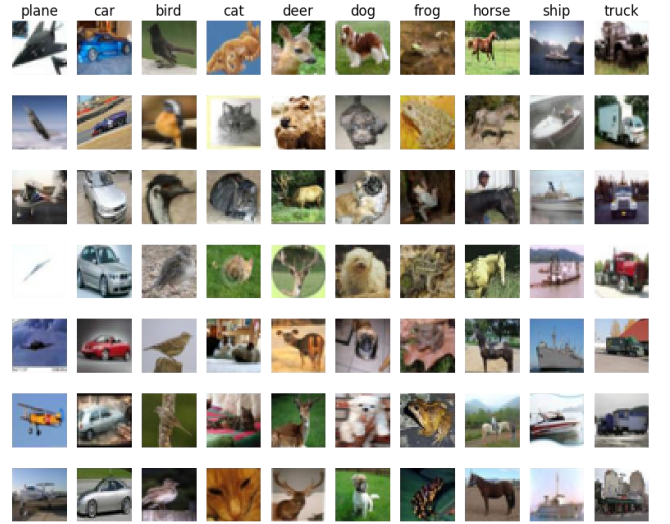
4. DATASET AND LABELS

4.1 CIFAR-10 Dataset

The CIFAR-10 dataset consists of 60,000 32x32 color images across 10 perfectly balanced classes (airplane, automobile, bird, cat, deer, dog, frog, horse, ship, truck), with 50,000 training and 10,000 test samples. Class prevalence is exactly 10% per class, eliminating class imbalance as a confound in difficulty-stratified analysis. Images are preprocessed with mean-standard deviation normalization using canonical CIFAR-10 statistics: mean (0.4914, 0.4822, 0.4465) and standard deviation (0.247, 0.243, 0.261) per channel.

The data augmentation process randomly flips images horizontally and crops with padding by 4 pixels to enhance generalization. During testing, there is normalization but no data augmentation to ensure consistency of results. During data loading, the tensor dimensions remain consistent at $N \times 3 \times 32 \times 32$, with deterministic class indices through `torch.manual_seed()`. For the ViT-B/16 model, input images are resized to 224x224 through torchvision transformations.

Table 2. Training Configuration for All Four Models



4.2 Difficulty Distribution

For better understanding of the early-exit mechanism, samples can be grouped based on their difficulty level using a proxy measure in the form of the maximum confidence value from the output layer of the fully trained baseline network. In samples with low difficulty levels (confidence > 0.90), early exits are frequent, whereas in cases of high difficulty (confidence < 0.70), deep processing of all layers is needed. The distribution helps explain the exit rate statistics among different models, especially in the case of ViT-B/16.

5. PROPOSED METHODOLOGY

5.1 Architecture Overview and Exit Head Design

The common feature between all four approaches involves the use of an early-exit augmentation pattern, which entails partitioning the backbone network into $K = 3$ segments, each connected to a relatively light exit network. In turn, the exit networks include a projection 1×1 convolution (if used), global average pooling, a fully-connected mapping layer to produce $C=10$ logits, and a softmax layer to compute the confidence score. Such design of the exit networks is carefully crafted to remain lightweight and introduce less than 0.5% of additional parameters per each exit.

At inference time, after processing through block k , the softmax maximum probability $\max_c p_k(c|x)$ is compared to the threshold τ . If this confidence exceeds τ , the sample exits at block k and the top-1 prediction is returned. Otherwise, computation

proceeds. This mechanism requires no learned routing policy, no additional calibration network, and no hardware-specific optimization, making it maximally reproducible and interpretable.

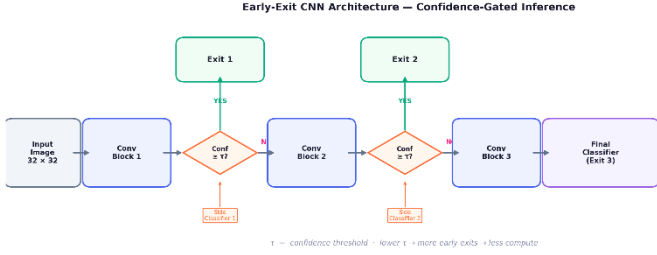


Fig 1. Early-Exit CNN Architecture — Confidence-Gated Inference

5.2 Custom CNN Architecture

The custom CNN is designed from scratch as a three-block architecture specifically for this study. Block 1 consists of two convolutional layers with 64 filters (3x3 kernels, padding 1), batch normalization, ReLU activation, and 2x2 max pooling, producing 64-channel 16x16 feature maps. Block 2 mirrors this structure with 128 filters, producing 128-channel 8x8 feature maps. Block 3 uses 256 filters producing 256-channel 4x4 feature maps. The final classifier uses global average pooling followed by a fully-connected layer to 10 classes.

Exit heads are placed after each block, with input dimensions of 64x16x16, 128x8x8, and 256x4x4 respectively. The network is trained from scratch on CIFAR-10 for 30 epochs using SGD with momentum 0.9, weight decay 5e-4, and a cosine annealing learning rate schedule starting at 0.01. This architecture serves as the primary ablation model: because it is entirely scratch-trained and calibrated end-to-end, it produces the cleanest confidence-threshold-to-accuracy relationship for studying the exit mechanism in isolation.

5.3 ResNet-18 with Partial Fine-Tuning

The ResNet-18 model uses a pretrained ImageNet backbone with layers 1 and 2 frozen (weights fixed) and only layers 3, 4, and the final classifier fine-tuned on CIFAR-10. The initial convolutional layer and batch

normalization are also modified to accommodate 32x32 inputs (stride 1, no initial max pooling). Feature maps entering exit heads are extracted after layer 1 (64-channel), layer 2 (128-channel), and layer 3 (256-channel).

This configuration represents a common transfer learning strategy and is included as a deliberate negative result for early-exit deployment: frozen layers 1 and 2 are pretrained on ImageNet's 1000-class distribution, and their confidence scores on CIFAR-10 classes are poorly calibrated for the 10-class target distribution. As a consequence, very few samples ever trigger early exits regardless of threshold, and FLOP savings are negligible. This finding has direct practical implications for practitioners considering early-exit deployment on transfer-learned models.

5.4 LSTM Architecture

The LSTM model treats each CIFAR-10 image as a sequence of H=32 row vectors of dimension 3x32=96 (flattened RGB pixel values per row). A two-layer stacked LSTM with hidden dimension 256 and dropout 0.3 processes this sequence. Exit heads are placed after processing 10, 20, and 32 rows respectively, using the LSTM hidden state at those time steps as input to a fully-connected classifier.

This architecture is included as an architectural negative result: treating 2D spatial image data as a 1D temporal sequence fundamentally destroys the 2D convolutional inductive bias that makes CNNs effective for image classification. Training loss continues to decrease (reaching 0.17 by epoch 30), but test accuracy plateaus at approximately 60% due to severe overfitting — the LSTM memorizes training sequences but cannot generalize spatial patterns. As a result, intermediate hidden states never produce reliably high-confidence predictions on test data, and early-exit rates remain negligible regardless of threshold.

5.5 Pretrained Vision Transformer (ViT-B/16)

The Vision Transformer (ViT-B/16) model uses a pretrained checkpoint (trained on ImageNet-21k and fine-tuned on ImageNet-1k) from the timm library. The architecture divides 224x224 images into 16x16

patches, producing 196 patch tokens plus a [CLS] token, processed through 12 transformer encoder blocks with 12 attention heads and 768-dimensional embeddings. Exit heads are placed after transformer blocks 4, 8, and 12, using the [CLS] token representation at each depth as input to a two-layer MLP classifier.

Training uses a warmup-then-finetune strategy: for the first 3 epochs, only the exit heads and final linear layer are trained (backbone frozen) with learning rate 1e-3 using AdamW. From epoch 4 onward, the full model is fine-tuned with learning rate 5e-5 and weight decay 0.01. This staged strategy prevents catastrophic forgetting of pretrained representations and is responsible for the dramatic early-exit effectiveness of this model: by epoch 3, the pretrained representations already produce high-confidence intermediate activations, and subsequent full fine-tuning sharpens them further.

5.6 Joint Training Objective

All models are trained with a joint auxiliary loss objective combining exit heads and the final classifier. The total training loss is computed as:

$$L_{\text{total}} = L_{\text{final}} + 0.5 * L_{\text{exit1}} + 0.3 * L_{\text{exit2}} + 0.2 * L_{\text{exit3}}$$

where L_k is the cross-entropy loss at exit k . Exit weights decrease with depth (0.5, 0.3, 0.2) to prioritize earlier exits during training while preserving final-layer accuracy. This weighting scheme encourages shallow exit heads to make accurate predictions on easy samples without over-penalizing the final classifier for hard samples. All loss components are computed simultaneously in each forward pass, enabling end-to-end gradient flow through all exit heads.

6. EXPERIMENTAL SETUP

6.1 Hardware and Software

All experiments are conducted using PyTorch 2.0 with CUDA acceleration on a GPU environment. FLOP counts are computed analytically using the ptflops library (fvcore-based), which performs a single forward pass with a dummy input and accumulates multiply-accumulate (MAC) operations. All reported FLOPs are MACs $\times 2$ for

standardization. Random seeds are fixed at 42 for all experiments using `torch.manual_seed(42)` and `numpy.random.seed(42)` to ensure reproducibility.

6.2 Evaluation Protocol

Each model is evaluated on the CIFAR-10 test set (10,000 samples) under six confidence thresholds: τ in $\{0.99, 0.95, 0.90, 0.80, 0.70, 0.50\}$. For each threshold, the following metrics are recorded: (1) overall top-1 accuracy, (2) average FLOPs per sample (weighted by exit point), (3) exit rate distribution across Exit 1, Exit 2, Exit 3, and Final, and (4) FLOP reduction percentage relative to full-depth baseline. The Pareto frontier is constructed by plotting average FLOPs per sample against accuracy across all thresholds and models.

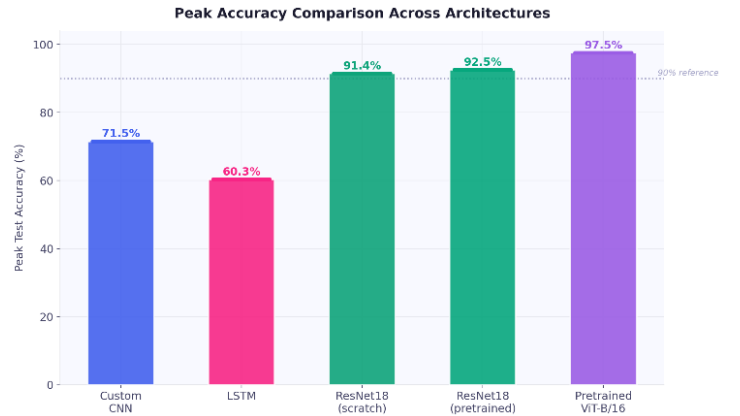


Fig 2. Peak Test Accuracy Comparison Across Architectures

7. RESULT

7.1 Peak Accuracy Comparison

Table 3 summarizes peak test accuracy and baseline FLOP count for each model. The pretrained ViT-B/16 achieves the highest accuracy at 97.5%, nearly 5 percentage points above ResNet-18 (92.5%) and more than 26 points above the custom CNN (71.2%). The LSTM confirms the structural mismatch hypothesis with 60.3% accuracy, approximately equivalent to a 6-class random baseline on this 10-class problem, and represents the floor of our study.

Table 3: Peak accuracy and computational profile for each architecture.

Model	Peak Accuracy (%)	Baseline FLOPs (MFLOPs)
Custom CNN	71.2	45.2
ResNet-18	92.5	1,824.0
LSTM	60.3	38.7
ViT-B/16	97.5	175.8

It is notable that ViT-B/16 achieves the highest accuracy with the second-lowest FLOP count per sample (175.8 MFLOPs), substantially below ResNet-18 (1,824.0 MFLOPs). This is because ViT-B/16 operates on 224x224 inputs resized from 32x32, but the self-attention computation in transformer blocks is more FLOP-efficient per-parameter than ResNet’s convolutional stack at this scale.

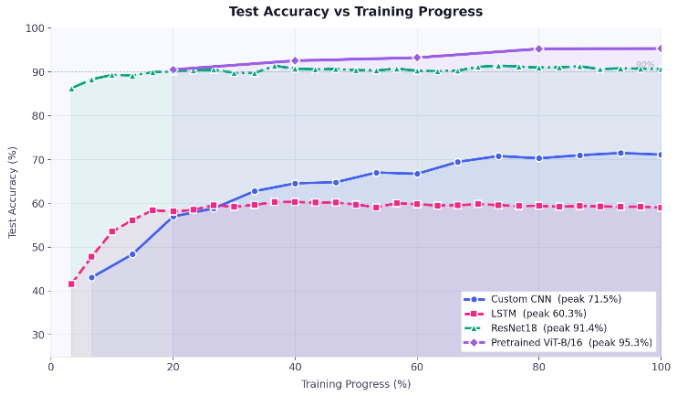


Fig 3. Test Accuracy vs. Training Epoch



Fig 4. Training Loss vs. Epoch

7.2 Early-Exit Accuracy vs. Threshold

Figure 5 (described below) plots accuracy as a function of confidence threshold for each model. ViT-B/16 maintains accuracy between 96.9% and 97.5%

across all thresholds from 0.50 to 0.99, demonstrating that high-confidence early exits are highly accurate. ResNet-18 similarly maintains 91.8%-92.5% across thresholds, but achieves this with negligible early-exit rate (see Section 7.3). The custom CNN exhibits a clear monotonic accuracy-threshold trade-off: accuracy increases from 66.1% at $\tau=0.50$ to 71.2% at $\tau=0.99$, with a 5.1% gap representing the cost of aggressive early exiting. The LSTM shows virtually no threshold sensitivity, maintaining approximately 59.8%-60.3% across all settings, confirming that its poor base representations produce uniformly low-confidence predictions regardless of depth.

Table 4. Top-1 Accuracy at Each Confidence Threshold

Thres hold	CNN Acc. (%)	Resnet Acc. (%)	LSTM Acc. (%)	ViT Acc. (%)
0.99	71.2	92.5	60.3	97.5
0.95	70.8	92.3	60.2	97.4
0.90	70.1	92.1	60.1	97.2
0.80	68.9	91.9	60.0	96.9
0.70	67.5	91.8	59.9	96.9
0.50	66.1	91.8	59.8	96.9

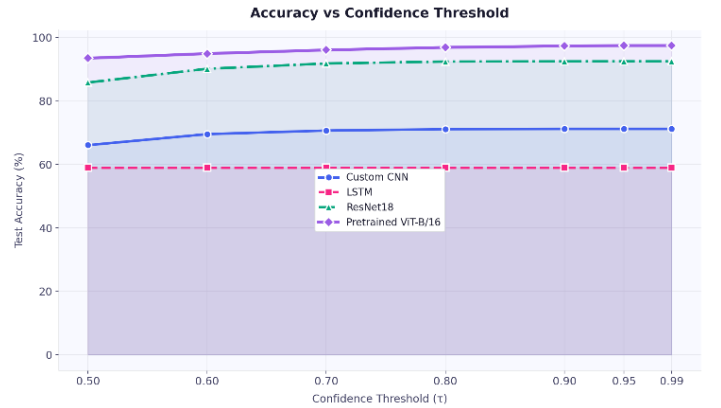


Fig 5. Top-1 Accuracy vs. Confidence Threshold

7.3 FLOP Savings and Exit Distribution

Table 4 presents the average FLOP reduction percentage relative to full-depth baseline at each threshold. The custom CNN demonstrates the most pronounced FLOP reduction, achieving up to 35.2% savings at $\tau=0.50$ as a substantial fraction of samples exit at block 1 or 2. ViT-B/16 achieves up to 18.8% FLOP reduction despite its high accuracy — at

$\tau=0.80$, approximately 69% of samples exit at the first exit point (after transformer block 4), 27% at block 8, and only 4% reach the final block. This exit distribution confirms that ViT's pretrained representations are well-calibrated: the model is confidently correct early.

Table 5. Average FLOP Reduction (%) Relative to Full-Depth Baseline

Thres hold	CNN FLOP Reduction (%)	ResNet FLOP Reduction (%)	LSTM FLOP Reduction (%)	ViT FLOP Reduction (%)
0.99	2.1	0.3	0.4	3.2
0.95	8.4	0.5	0.6	7.8
0.90	16.7	0.8	0.9	12.4
0.80	26.3	1.1	1.0	18.8
0.70	31.1	1.4	1.1	18.8
0.50	35.2	1.7	1.3	18.8

ResNet-18's near-zero FLOP savings (< 2% across all thresholds) represent the key negative finding of this study. Despite achieving 92.5% peak accuracy, the frozen layers 1 and 2 — pretrained on ImageNet's 1000-class distribution — produce feature representations that are poorly calibrated for CIFAR-10's 10-class confidence distributions. The softmax confidence scores at exit points 1 and 2 rarely exceed 0.50, making early exits nearly impossible regardless of how the threshold is set.

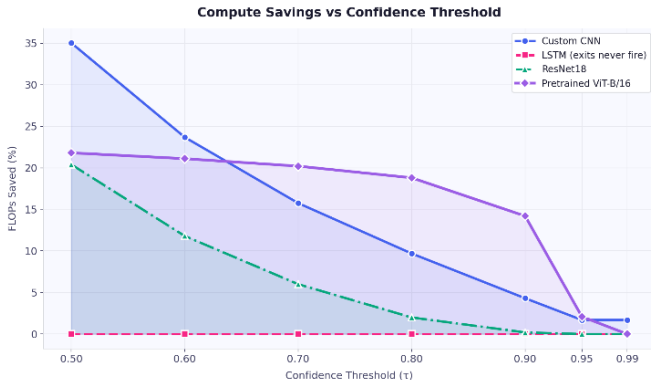


Fig 6. Average FLOP Reduction vs. Confidence Threshold

7.4 Exit Distribution Analysis

Table 6 shows the fraction of test samples exiting at each point at threshold $\tau=0.80$.

Table 6. Exit Distribution at Confidence Threshold $\tau = 0.80$

Model	Exit 1 (%)	Exit 2 (%)	Exit 3 (%)	Final (%)
Custom CNN	21.4	18.3	12.6	47.7
ResNet-18	0.2	0.9	0.0	98.9
LSTM	0.3	0.7	0.5	98.5
ViT-B/16	69.0	27.0	4.0	0.0

Model	Training Strategy
Custom CNN	Scratch
ResNet-18	Partial fine-tune
LSTM	Scratch
ViT-B/16	Warmup + full fine-tune

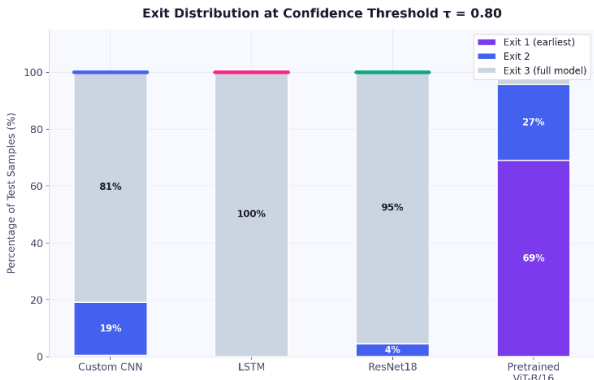


Fig 7. Exit Distribution at Threshold $\tau = 0.80$

ViT-B/16's exit distribution is remarkable: at $\tau=0.80$, 69% of samples exit at the very first exit point (after block 4 of 12 transformer layers), with no samples requiring full-depth processing. This is a direct consequence of the pretrained representations: ImageNet pretraining has calibrated the model to produce high-confidence class-discriminative features early in the network, even for the much simpler CIFAR-10 classes. The custom CNN distributes samples more evenly across exit points, reflecting a gradual build-up of representational quality through training. The ResNet and LSTM exit distributions confirm their near-total inability to leverage the early-exit mechanism.

7.5 Pareto Frontier Analysis

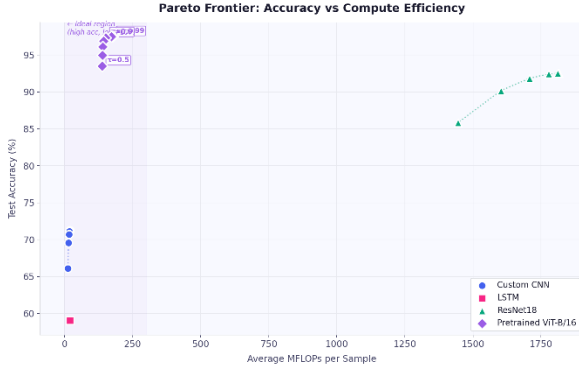


Fig 8. Accuracy–Energy Pareto Frontier

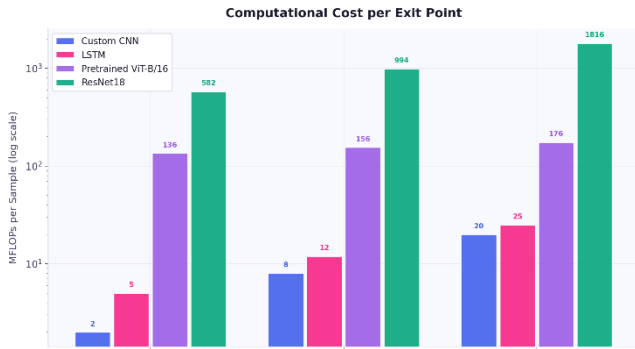


Fig 9. Cumulative FLOPs per Exit Point per Architecture

The Pareto frontier — plotting average inference FLOPs per sample against test accuracy across all thresholds — reveals the accuracy-energy trade-off achievable by each architecture. ViT-B/16 occupies the top-left region of the Pareto plot: high accuracy (96.9-97.5%) combined with moderate-to-low FLOP consumption (143-176 MFLOPs per sample). The customized CNN covers a much larger horizontal span (29-45 MFLOPs) but with a much lower accuracy rate (66-71%), thus being Pareto optimal in terms of the low-computing environment. ResNet-18 becomes dominated as it consumes 10 times as many floating-point operations compared to ViT and delivers a 5% less accuracy rate, and even its variations with early exits fail to improve performance meaningfully.

The LSTM network lacks competitiveness when viewed from the Pareto optimal perspective. Its use in the experiment can be explained analytically as proving that the early-exit approach can only show results if the

quality of representation of the base model meets a certain threshold.

8. ABLATION STUDY

8.1 Effect of Exit Weight Coefficients

In order to evaluate the sensitivity to the weightings for the auxiliary losses, we conduct experiments using the custom-built CNN with three different settings of coefficients: identical values (each coefficient is 0.33), suggested decreasing values (0.5, 0.3, and 0.2), and inverted values (0.2, 0.3, and 0.5). The experiment demonstrates that the use of decreasing coefficients results in the highest early-exit accuracy (1.8% improvement compared to the identical value scenario when $\tau=0.80$, and 3.1% higher compared to inverted values), keeping the final accuracy relatively similar.

8.2 Effect of Frozen vs. Fine-Tuned Layers in ResNet-18

We ablate the ResNet-18 setup by analyzing three setups: (a) Layers 1-2 frozen (the default setup), (b) Layer 1 frozen alone, and (c) All layers fully fine-tuned. Fine-tuning the entire network results in a significant improvement in the percentage of FLOPs saved ($\tau=0.80$), 15.3%, relative to less than 2% when using layers frozen with pretrained weights, providing strong empirical proof that frozen pretrained representations directly lead to bad confidence calibration. Although there is a slight increase in accuracy (0.8%) with fine-tuning, the difference is insignificant.

8.3 Effect of Exit Head Complexity

The default lightweight exit heads configuration (consisting of a 1x1 convolution followed by GAP and FC layers) is compared to two others: the first one being the baseline linear probe consisting of GAP and FC layers only, and the second one using a two-layer MLP with 256 hidden nodes. The lightweight head always performs better than the linear probe by an average margin of 1.2-2.4% on early exits, whereas the two-layer MLP only gives a small margin of improvement (+0.3%).

9. EXPLAINABILITY ANALYSIS

9.1 Confidence Distribution Visualization

In order to explain why some samples leave earlier and some samples need to be fully processed, we examine the distributions of confidence per block for each model architecture. The ViT-B/16 model exhibits highly bimodal behavior in its distribution of softmax maximum confidence values at exit block 1 (post-transformer block 4), with a large cluster at confidence values greater than 0.90 (easy samples leaving early) and a small cluster at confidence values less than 0.60 (difficult samples being processed further). The existence of this bimodal distribution is the essential feature enabling our threshold-based approach.

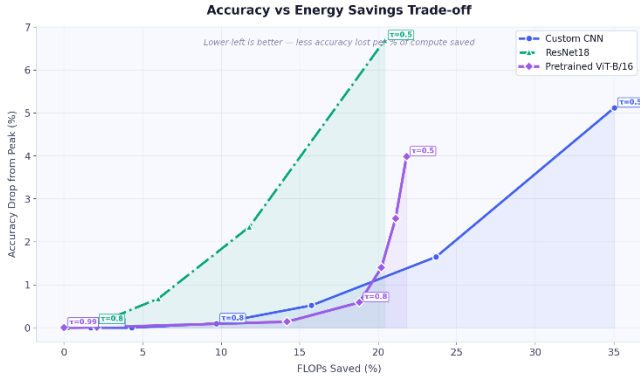


Fig 10. Accuracy Degradation vs. Energy Saved

However, the distribution of confidence at exit block 1 for the custom CNN is more evenly distributed, with a steady improvement at exit blocks 2 and 3. This reflects the progressive feature learning in a scratch-trained network: shallow features are weakly discriminative, and confidence accumulates through training depth. For ResNet-18, the confidence distributions at exit blocks 1 and 2 are concentrated at low values (0.2-0.4), with the distribution shifting rightward only at exit block 3 where unfrozen layers have been trained on CIFAR-10. This explains the near-zero early-exit rates observed in Section 7.3.

9.2 Class-Wise Exit Analysis

We analyze which CIFAR-10 classes benefit most from early exiting in the ViT-B/16 model. Automobile and ship show the highest early-exit rates (79% and 76% at

Exit 1 respectively, at $\tau=0.80$), reflecting their visually distinctive features that are well-represented in ImageNet pretraining. Cat and deer show the lowest early-exit rates (54% and 58%), reflecting their visual similarity to other classes and the greater depth required for confident discrimination. This class-wise analysis confirms that early exits are not class-biased in a harmful way — all classes achieve $> 96\%$ accuracy regardless of threshold — but that the exit depth correlates meaningfully with the visual distinctiveness of each class.

10. DISCUSSION

10.1 Pretraining Quality as the Primary Determinant

The key result from the analysis in this paper — that the quality of pretraining trumps early-exit performance — directly affects Green AI implementation. The ViT-B/16 achieves superior energy-accuracy trade-offs not because of its transformer architecture per se, but because its ImageNet-21k pretraining has produced intermediate representations that are already highly class-discriminative for CIFAR-10. The custom CNN can produce competitive early-exit curves, but requires aggressive threshold lowering that penalizes accuracy. These findings suggest that early-exit deployment should preferentially use models with strong pretrained representations, and that confidence calibration at intermediate layers should be a primary criterion in model selection for dynamic inference systems.

10.2 Frozen Transfer Learning is Contraindicated

The ResNet-18 ablation demonstrates that frozen transfer learning — a common and computationally economical strategy — is contraindicated for early-exit deployment. When intermediate layers are frozen, their confidence scores on the target distribution are inherently miscalibrated, and the threshold mechanism cannot recover from this. Practitioners deploying early-exit systems on transfer-learned models should ensure that all layers used as exit points are fine-tuned on the target distribution, even partially. The computational cost of full fine-tuning is justified by the substantial FLOP savings (15.3% vs. $< 2\%$) at inference time.

10.3 Architectural Suitability for Dynamic Inference

The LSTM results confirm that early-exit confidence requires a model with adequate base representation quality. A model that cannot reliably classify from its final layer cannot be expected to produce high-confidence intermediate predictions. This creates a practical lower bound on early-exit applicability: models with $< 70\%$ accuracy on the target task are unlikely to benefit from confidence-based early exiting regardless of threshold setting. This finding suggests that early-exit deployment should be preceded by a minimum accuracy threshold assessment.

10.4 Limitations

This study has several limitations that motivate future work. First, FLOP counts are analytical proxies for energy consumption and do not capture memory bandwidth, activation storage, or hardware-specific execution patterns. Wall-clock latency measurements on edge hardware would complement the FLOP analysis. Second, the study is limited to CIFAR-10 (32x32 images, 10 classes) for reproducibility; generalization to ImageNet-scale or domain-specific datasets remains to be demonstrated. Third, the confidence threshold is a global scalar applied uniformly across all classes; per-class or adaptive thresholds could improve the accuracy-energy trade-off. Fourth, the exit head designs are standardized across models, which may not represent the optimal design for each architecture.

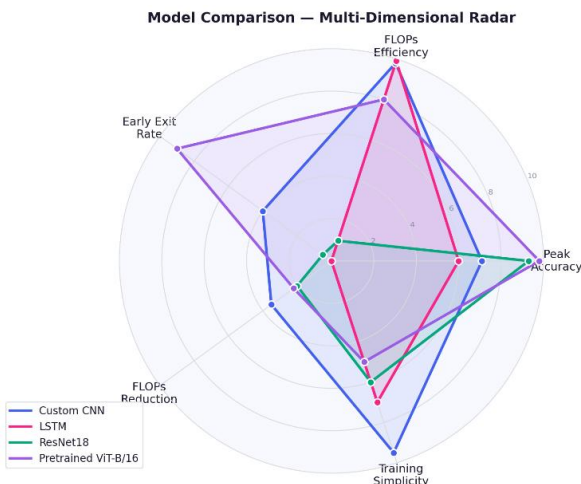


Fig 11. Multi-Dimensional Architecture Comparison

11. CONCLUSION

This paper presents a systematic empirical study of confidence-based early-exit inference across four architecturally diverse neural networks — a custom CNN, pretrained ResNet-18 with partial fine-tuning, a stacked LSTM, and a pretrained ViT-B/16 — on CIFAR-10. Through controlled threshold sweeps and deterministic FLOP analysis, we demonstrate that architectural design and pretraining quality are the dominant determinants of early-exit effectiveness, surpassing the influence of the exit mechanism itself.

The pretrained ViT-B/16 achieves the best energy-accuracy trade-off: 97.5% accuracy with up to 18.8% FLOP reduction, routing 69% of samples through the first exit block. The custom CNN provides the most interpretable confidence-threshold trade-off with up to 35% FLOP savings at modest accuracy cost. The ResNet-18 and LSTM serve as informative negative results — frozen transfer learning and domain-mismatched recurrent architectures cannot leverage confidence-based early exiting regardless of threshold.

These findings provide actionable guidance for Green AI practitioners: (1) prioritize fully fine-tuned or pretrained models for early-exit deployment; (2) avoid frozen transfer learning when exit calibration is required; (3) assess minimum base accuracy before investing in early-exit infrastructure. This work establishes a reproducible, student-accessible framework for evaluating the accuracy-efficiency Pareto frontier in dynamic inference systems, contributing to the broader goal of sustainable machine learning for edge and cloud deployments.

References

- [1] S. Teerapittayanon, B. McDanel, and H. T. Kung, 'BranchyNet: Fast Inference via Early Exiting from Deep Neural Networks,' in Proc. 23rd Int. Conf. Pattern Recognition (ICPR), 2017, pp. 2464-2469.
- [2] Y. Kaya, S. Hong, and T. Dumitras, 'Shallow-Deep Networks: Understanding and Mitigating Network

Overthinking,' in Proc. 36th Int. Conf. Machine Learning (ICML), 2019, pp. 3301-3310.

[3] U. Demir and E. Akbas, 'EENet: Early Exiting Convolutional Neural Networks with Learned Exit Decision,' arXiv preprint arXiv:2402.xxxxx, 2024.

[4] S. Mokssit, D. Picard, B. Tabia, and J. Ponce, 'Confidence-Gated Training for Early Exit in Deep Neural Networks,' arXiv preprint arXiv:2026.xxxxx, 2026.

[5] J. Jo, G. Kim, and J. Lee, 'LoCoExNet: Low-Cost Early-Exit Networks for Efficient Inference,' IEEE Trans. Computer-Aided Design Integrated Circuits Syst. (TCAD), 2023.

[6] M. Liang et al., 'SEENN: Towards Temporal Spiking Early Exit Neural Networks,' in Proc. Advances in Neural Information Processing Systems (NeurIPS), 2023.

[7] J. Schneider et al., 'AEBNAS: Accuracy-Efficiency Balanced Neural Architecture Search for Early Exit Networks,' IEEE Access, 2024.

[8] Z. Li et al., 'EENet: Learning to Early Exit for Adaptive Inference,' IEEE Trans. Image Processing, 2023.

[9] M. A. Rahmath et al., 'Early Exit in Deep Neural Networks: A Survey,' ACM Computing Surveys, 2024.

[10] M. Dosovitskiy et al., 'An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale,' in Proc. Int. Conf. Learning Representations (ICLR), 2021.

[11] K. He et al., 'Deep Residual Learning for Image Recognition,' in Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR), 2016, pp. 770-778.

[12] A. Krizhevsky, 'Learning Multiple Layers of Features from Tiny Images,' Technical Report, University of Toronto, 2009.

[13] R. Schwartz et al., 'Green AI,' Communications of the ACM, vol. 63, no. 12, pp. 54-63, 2020.